Department of Artificial Intelligence and Machine Learning

Session Plan with CO mapping to each lecture

Format 5

Symbiosis Institute of Technology, Pune						
Session Plan						
Name of the department – Artificial Intelligence & Machine Learning						
Name of the course- Supervised Machine Learning (0701260304)					**Credit – 4 (3+1)**	
Semester - 3					**Batch - 2025-29**	
Name of the faculty- Dr. Ketan Kotecha, Dr. Sumit Kumar, Dr. Rahesha Mulla, Dr. Ananya Bhattacharjee						
Lect. No	**Unit No.**	**Points to cover**	**Methodology**	**Faculty Conducting**	**Lecture/ Exp. Learning/ Evaluation**	**CO**
1	1	Definition, evolution, and need of Machine Learning	Lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
2	1	Supervised, Unsupervised, and Reinforcement Learning	Theory lecture, Presentation, and case study discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
3	1	Classification and Regression with Examples	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
4	1	Clustering and Association techniques with applications	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
5	1	Basic concepts of Reinforcement Learning, agents, rewards, and applications	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
6	1	Applications of Machine Learning in healthcare, finance, retail, automation, and cybersecurity	Group discussion on a real-world case study	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO1
7	1	Importance of data, data collection, and preprocessing	Theory lecture, Presentation, and dataset demonstration	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1

8	1	Training, Validation, and Test datasets	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
9	1	K-Fold Cross Validation, Hold-Out method, and Leave-One-Out method	Lecture with interaction /Video	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO1
10	1	Feature extraction, transformation, and encoding techniques,	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
11	1	Filter, Wrapper, and Embedded feature selection methods	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
12	1	Applications of Supervised Machine Learning and revision	Theory lecture, Presentation, and revision	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO1
13	1	CA 1 (course certification & report)	NPTEL-SWAYAM/ Coursera/ Udemy/Infosys Springboard course completion certificate, & report submission	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Evaluation	CO1
14	2	Difference between Regression and Classification	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
16	2	Binary and Multiclass Classification concepts	Theory lecture with examples, Presentation, and revision	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
17	2	Introduction to Linear Classifiers, Logistic Regression: concepts and applications	Theory lecture with examples, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
18	2	Support Vector Machines (SVM): concepts and working	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
19	2	K-Nearest Neighbors (KNN): concepts and applications	Theory lecture with problem solving, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
20	2	Bayes Theorem and Naive Bayes Classifier	Theory lecture with problem solving, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2

21	2	Model Evaluation and Comparison Techniques	Theory lecture and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
22	2	Cross-Validation, Bootstrapping, Leave-One-Out, and Hold-Out methods	Lecture with examples, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO2
23	2	Classification Techniques	An analytical discussion of different classification algorithms	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO2
24	2	Confusion Matrix and performance analysis, precision, Recall, and F1 Score, ROC, and AUC evaluation metrics	Lecture with interaction and problem-solving	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO2
25	1,2	CA 2 (Unit Test)	Pen-and-Paper Test	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Evaluation	CO1, CO2
26	1, 2	CA 2 (Unit Test)	Pen-and-Paper Test	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Evaluation	CO1, CO2
27	1, 2	CA 2 (Unit Test)	Pen-and-Paper Test	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Evaluation	CO1, CO2
28	3	Dependent and Independent Variables, Correlation and Covariance concepts	Theory lecture with numerical examples, presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
29	3	Ordinary Least Squares (OLS) method	Lecture based on problem-solving with interaction	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO3
30	3	Simple Linear Regression, Multiple Linear Regression	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
31	3	Polynomial Regression	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
32	3	Regression Techniques	Analytical discussion and interpretation of regression models	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO3

33	3	Gradient Descent optimization technique	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
34	3	Evaluation metrics: R-squared and Mean Squared Error, Additional evaluation metrics for regression models	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
35	3	Underfitting and Overfitting Concepts, Bias-Variance Trade-off	Theory lecture, Presentation, and discussion/video	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO3
36	3	Regularization techniques: Ridge and Lasso Regression	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO3
37	4	Introduction to Decision Tree Classifier and basic terminology	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4
38	4	Decision Tree construction and splitting criteria	Theory lecture with numerical examples, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4
39	4	Pruning Techniques in Decision Trees	Theory lecture, Presentation, and discussion / Video	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO4
40	4	Concept of Ensemble Learning Methods	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4
41	4	Random Forest algorithm and working principle	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4
42	4	Bagging technique and its applications, Boosting technique and its importance	Theory lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4
43	4	Ensemble Learning Techniques	Discussion on comparative study and use cases/ Video	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Experiential Learning	CO4
44	4	ADA Boost algorithm and working, Gradient Boosting technique	Lecture, Presentation, and discussion	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Lecture	CO4

45	4	CA 3 (Case Study Assignment)	Real-world case study based on recent literature & report submission	Dr. Sumit Kumar/ Dr. Rahesha Mulla/ Dr. Ananya Bhattacharjee/ Dr. Ketan Kotecha	Evaluation	CO3, CO4

Signature of Course Incharges:







Dr. Sumit Kumar Dr. Rahesha Mulla Dr. Ananya Bhattacharjee



Signature of HoD